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May 18, 2020

Mr. Ryan Schmit, RA  
ThenDesign Architecture  
4135 Erie Street  
Willoughby, Ohio 44094

Re: Floor Slab Investigation  
High School  
Brecksville Broadview Heights School District  
IAL No. 20053A

Dear Mr. Schmit:

As authorized by you, Isaac Lewin PE has reviewed the floor slab conditions at the High School for the Brecksville Broadview Heights School District Campus in Brecksville, Ohio. The review was requested due to the continued movement of the floor slab. The school was visited on February 4, 2020, to observe the existing conditions. The site visit was limited to visual observations of existing conditions. Geotechnical testing of the soils was requested and performed at a later date. Photographs of the existing conditions were taken and are on file at this office. Selected photographs are attached to this report.

The investigation, in this writer's professional opinion, found the floor slabs in Units "D" and "E" are continuing to settle, resulting in damaged partition walls, masonry walls and masonry columns enclosures. The fill used to construct the building appears not to have been compacted properly and is considered soft. The slab settlement also probably influences the cracking of the masonry on the first floor since the masonry column enclosures are supported on the slab rather than being supported on the column footing. The issues observed do not appear to affect the structural safety of the building, since the main structural framing elements are the steel frame, not the masonry. However, the fact that they have continued is a maintenance and public perception issue.

The following documents were used in preparation of this report:

- Fanning Howey Drawings RA 1.13, RA 1.14, RA 1.27, RA 1.28, RA 1.29, S 1.7, and RG 2.2
- Timmerman Geotechnical Group Inc. "Floor Evaluation Report" TGG #067059A, dated July 10, 2008
- ECS Solar Testing Laboratories Report, No. S020189, dated April 10, 2020
- I A Lewin, PE and Associates Report, Project No. 10496, dated June 1, 2011

The existing high school was designed and constructed between 1995 and 1997. The building was designed by Fanning Howey Associates Inc. of Columbus, Ohio and Geary Moore and Ahrens, Inc of Brecksville, Ohio. The area of the building that is the subject of this examination is known as Unit "D" and "E" per the construction documents. For purposes of this report the orientation of the long axis of the building is north-south with the gymnasium is considered the north end of the building and

the auditorium is the south end. Based on this Units "D" and "E" are considered the middle south of the building. This part of the building is a two-story structure with a partial basement on the west side of the building. The foundation wall is a substantial cast in place cantilever style concrete retaining wall. The building uses a superstructure of steel framing with moment connections for lateral bracing. The floors are concrete deck on steel joists and beams. The roof is steel deck on steel joists and beams. The structural steel is surrounded by masonry construction. The masonry surrounds are supported on the floor slabs. The corridor walls and some classroom divider walls are masonry also. Most of the classroom partition walls are of gypsum wallboard construction. The exterior walls consist of an outer wythe of brick, air space and an in wythe of concrete masonry. The building's foundations are shallow conventional reinforced concrete footings. It was noted the site was raised from 5 feet to 15 feet with compacted fill. The ground floor slabs are cast in place concrete placed on the subgrade. This is as recommended by the PSI report.

Based on the correspondence, the slab movement issues developed about three years after completion of the building. According to the Atlas Pier correspondence the slab problems began appearing in Rooms 165 through 167, which is in the Unit "E" portion of the building. Atlas piers installed a series of "jack" piles under the perimeter foundation wall and the interior column footings. Atlas Piers also injected a slurry material under the floors in an attempt to level the floors and prevent further settlement of the slabs. Based on the correspondence available, no work has been performed since 2008.

The Timmerman Geotechnical (TGG) report found the soils under the floors to be soft soils. They performed one test in Section "D" in Room 144 and three tests in Unit "E", Rooms 165, 166 and 175. They consistently found depths of fill commensurate with the grading plan, RG 2.2. However, they found the soils to be soft for most of the fill depth.

The Solar Testing Laboratory (STL) investigation of 2020 found similar soil issues to that found in the 2008 TGG Report. The TGG report stated they use a Wildcat Penetrometer to establish the soil conditions. Their results indicated varying densities of the soils (very loose to loose for sandy type soils and soft to medium stiff for clay type soils). Two of the three STL boring were located in the same room as TGG's and the third one was located in an adjacent room. STL took actual boring where they were able to obtain actual soil samples as well as use the testing to determine the soils characteristics. The STL results are essentially identical to TGG's. STL obtained sufficient samples to perform two tests at their laboratory. One test was percentage of water in the soil and the relative density of the soil. They also performed a laboratory density test. In addition to finding the soils were the same, they also found that the in-place water content of the existing soils to be greater than the preferred water content of the laboratory compaction test. This indicates one of two issues. The first, in my opinion, is the original fill was not placed the preferred water content but was higher than the preferred. This higher water content led to the settlement issues observed for 20 years. The second, as described by STL, the soils were compacted too dry and this caused the soils to absorb too much water during construction resulting also in the conditions observed for years. In either case, this needs to be corrected. STL also noted voids under the slab. The voids are small enough to raise questions about the slab settlement, however the slabs have settled which reduces the void between the slab and fill and in places the slab was "mudjacked" which also filled in the voids. So, while the voids could have been larger, the lack of voids is easily explained.

In late 2010, I A Lewin PE and Associates was authorized to investigate the on going slab settlement in the Units D and E of the building. An extensive documentation of the existing conditions at the time was made and a report prepared. A copy of that report is attached to this report. We found that the slabs had settled anywhere between ¼ inch up to two inches and the slabs were moving so much

that the control joints, usually and 1/8 to 1/4-inch-wide, were open as much as 1 inch. Also, in many rooms on both the first and second floors any masonry walls that were supported on the concrete floor slabs developed cracks. Some were very significant.

As authorized by TDA, this writer performed a limited visual survey of the damages on February 4, 2020. The survey started at the north end of Unit "D". Settlement issues, similar to those observed in 2010, were observed on the first floor, including drywall cracking, masonry cracking, and slab settlement. Multiple changes were noted including room number, replacement of partition walls and one loose concrete masonry unit was replaced. At this site visit, geotechnical testing was requested and authorization was given to obtain quotes. Solar Testing Laboratories (STL) was selected and their testing was performed. As noted in a previous paragraph, STL determined the same soils conditions were present, matched what TGG

Based on these current and past observations there is a correlation between the slab settlements and the presence of fill (as referenced in the TGG and STL reports). The TGG report found the soils, for the full depth of the fill, to be inadequately compacted. In reviewing the problems, no signs of building settlement of the exterior walls or at interior columns were observed. The exterior brick veneer shows no signs of unusual cracks or settlement. However, it was noted in 2010 that the exterior walls had been underpinned by Atlas Piers with success. The only indications of soil issues are the slab settlements in both Units "D" and "E".

Based on this review of the existing conditions, design drawings, past investigations, it is this writer's professional opinion slab settlement issue is directly due to the improper placement or compaction of the fill soils. Although the building foundations do not appear to be settling, the slabs continuing to settle. This has been ongoing problem since 2000, after the school opened. The past attempts to mudjack (pressure grout) the voids under the slabs have failed since the slabs continue to settle, as have the random attempts to underpin the slab by installing pin piles. Part of the problem with the previous use of the mini-piles is the slab is supported by the grade. It is not designed to be a structural slab that could span between piles or the piles would have to be spaced no more than 6 feet apart to be effective. This was not the case. It must be stated, based on the evidence observed and the analysis presented, there appears no structural safety issues with the building.

To correct the slab settlement issues, the following suggestions are offered:

#### Option 1 – Permanent Repair

The slabs should be replaced with reinforced structural concrete slabs supported by mini-piles spaced at acceptable distances using a 6-inch reinforced concrete slab. Replacement of the slabs will cause a lengthy shutdown of the rooms and construction issues for a building in use. The structural slabs will need to be isolated from the building columns to avoid adding any loads from the ground floor to the columns so as to not to overload the column foundations. Masonry wall and drywall partition repairs will need to be considered. Also, any underground piping in these rooms will need to have hangers added so that should the soil continue to settle, the piping will not be damaged and function properly.

#### Option 2 – Temporary Repair

The slabs can be mudjacked to the greatest extent possible using expansive polyurethane foam. This product is now commercially available and is accepted by Ohio Department of Transportation. Per STL's report they anticipate that the greatest amount of soil consolidation and settlement has taken place. This is born out by the fact STL reported the void between the slab is small. This could be due to the settled slab resting on the sunken soil or the tests were taken at location that had been previously mudjacked. This option is a viable option and will cost substantially less than replacing

the slab as recommended in Option 1. There is still the possibility that there will be future settlement of the slab should the soil settlement continue. However, the incremental cost for mudjacking sometime in the future will still be less than the slab replacement option.

Based on this analysis, should the School Board wish a permanent, long term solution, slab replacement with a helical piles and structural slab is the preferred one. If costs are an issue, then mudjacking the slab per Option 2 is recommended with the knowledge that additional mudjacking will probably be needed in the future.

Attached to this letter, are copies of original building drawings RA 1.13 and RA 1.14, first floor plans showing the TGG and STL test locations, and selected photographs of existing conditions from 2010 and 2020.

This letter is based on a limited examination of an existing, finished structure. No warranty is made or implied that all defects were observed and reported. Due to the finished construction, concealed conditions may exist that would affect this report; this office assumes no liability for concealed conditions that may affect the extent or costs of any repair work needed. If conditions assumed by this office are found to be different than those described in this letter, this office should be notified. This office reserves the right to adjust the information presented in this letter should additional information become available. This office does not assume any responsibility for any defects in design or construction, whether observed or not, since this office was not involved in the original design and construction of this building. The opinions, conclusions and recommendations contained in this letter are based on this writer's judgment and experience as a practicing Structural Engineer.

Please feel free to contact us with any questions or if we can be of further assistance in resolving these issues.

Sincerely,  
**I A LEWIN, PE AND ASSOCIATES**

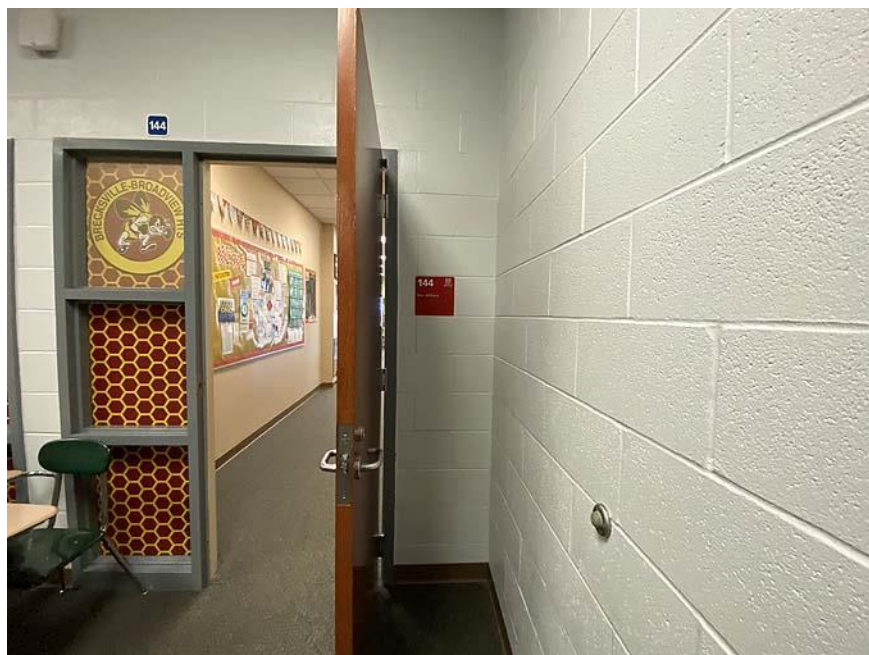
Isaac Lewin, PE

File  
Attachments

## PHOTOGRAPHS

### 2020 Photographs Using Current Room Numbers Room Number followed by Typical Interior Condition









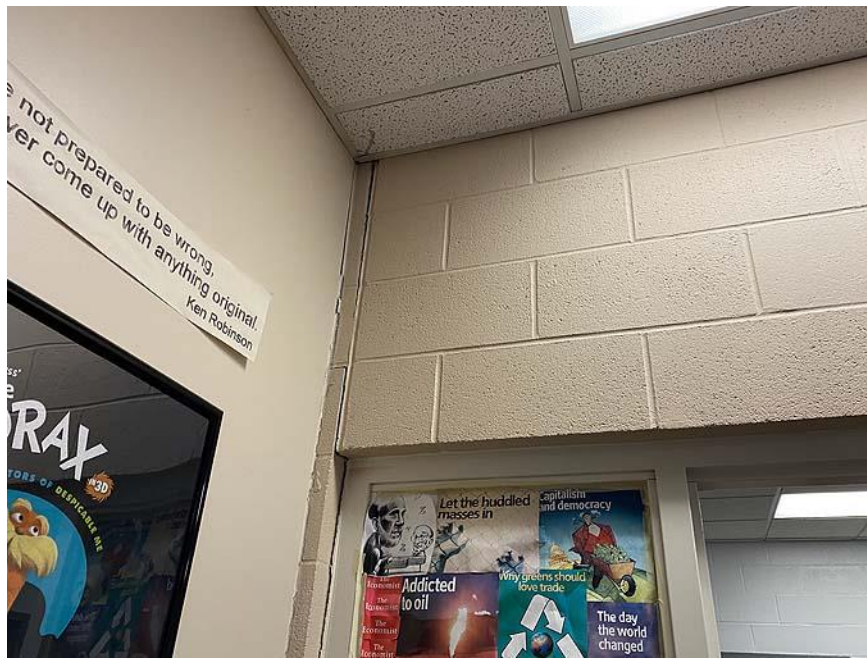


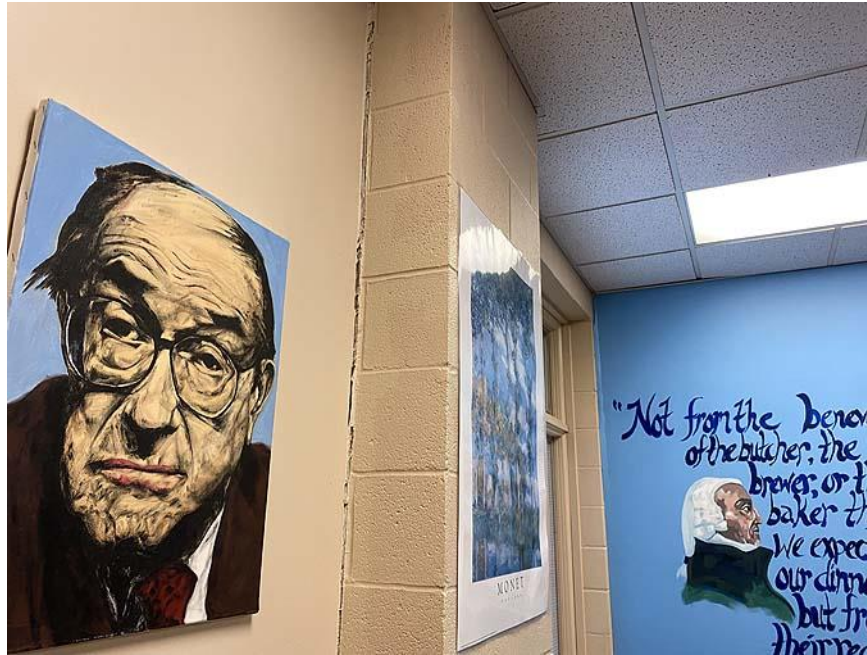






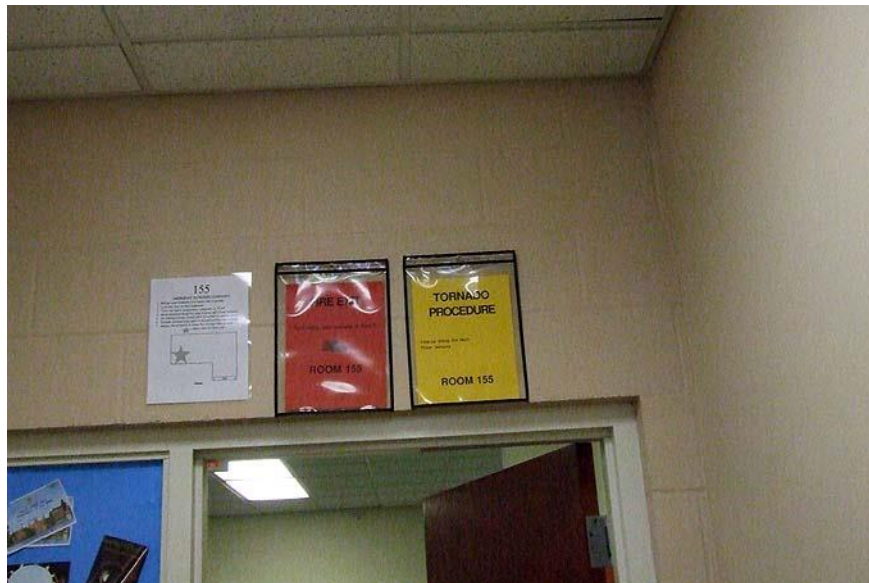




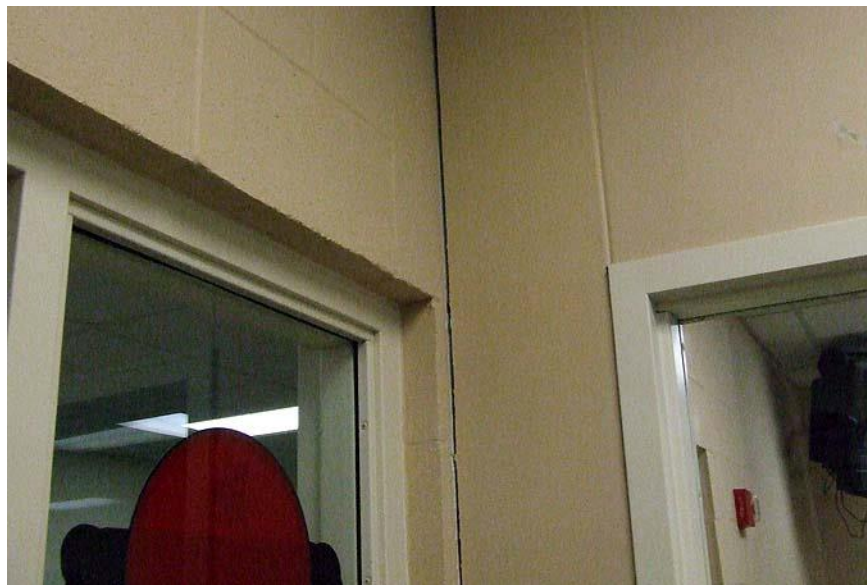


**2010 Photographs Using Past Room Numbers  
Room Number followed by Typical Interior Condition**





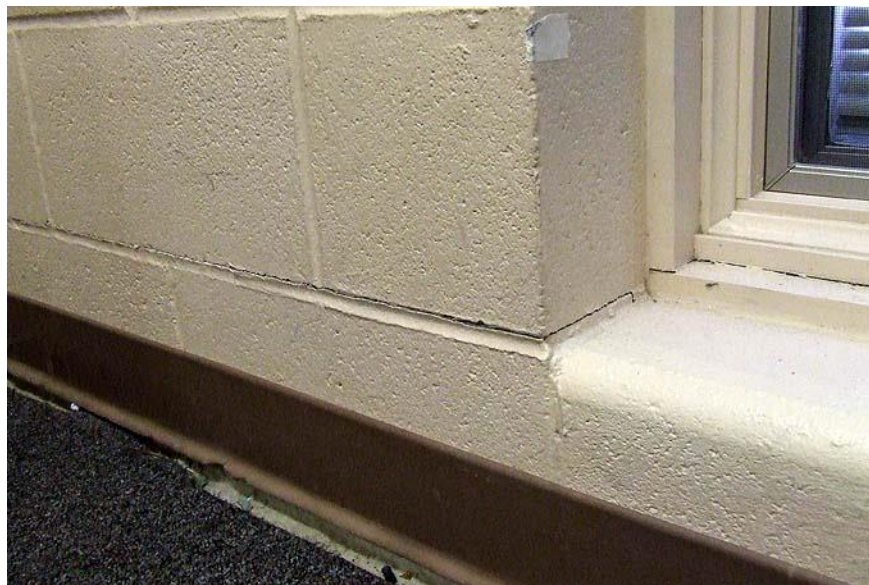












For the following photographs, page 13 for current conditions

